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How is accuracy in Analysis determined?

Learn how CAPS2 impacts your accuracy score on Chess.com and understand how the system works.

Updated over 10 months ago

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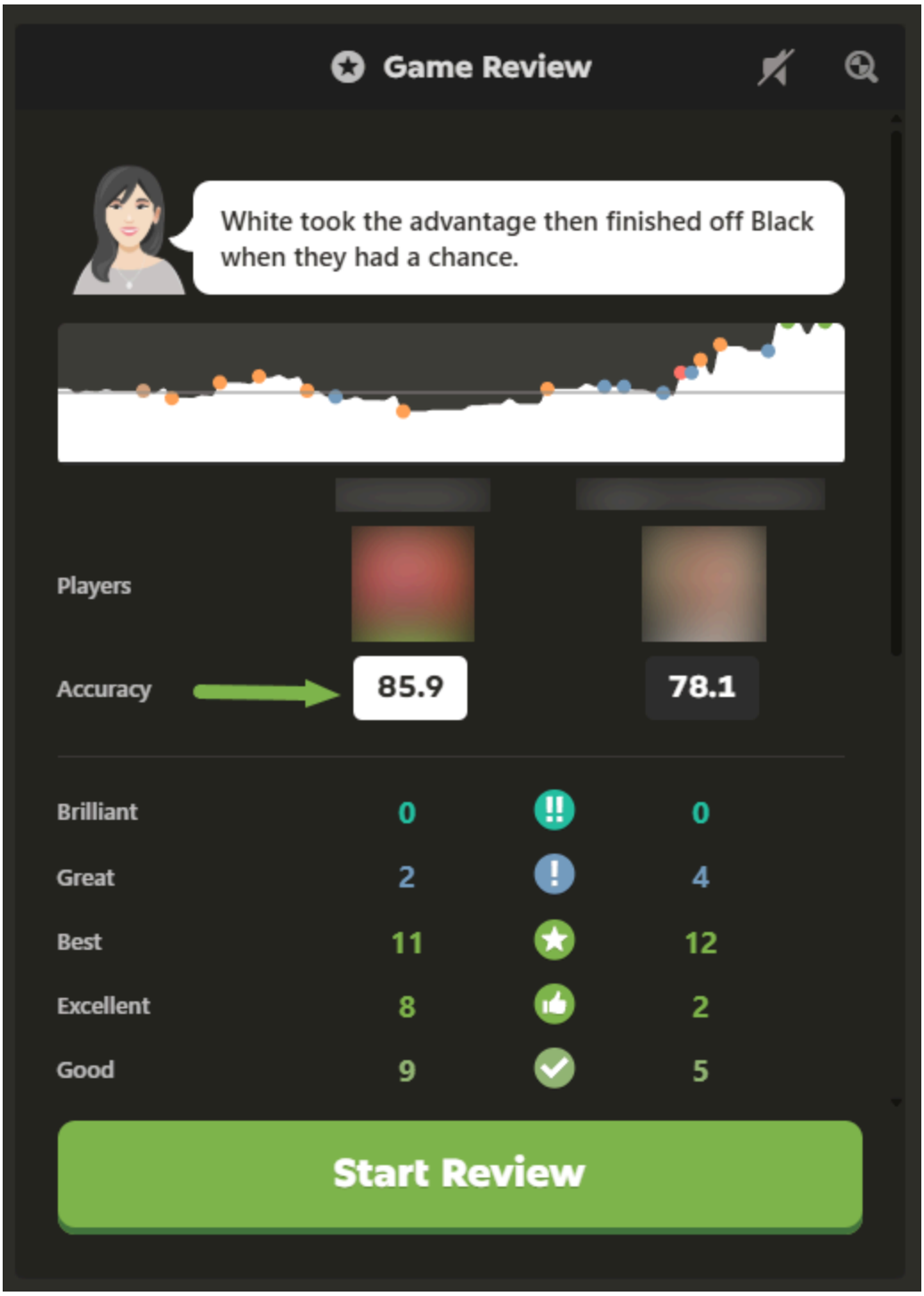
Each game review shows you an accuracy score and classifies all of your moves. But how are the accuracy and classifications decided?

Accuracy scores

Accuracy is a measurement of how closely you played to what the computer has determined to be the best possible play against your opponent's specific moves. The closer you are to 100, the closer you are to 'perfect' play, as determined by the engine.



CAPS and CAPS2

Chess.com's Accuracy score is now powered by 'CAPS2,' an improved version of the original Chess.com 'CAPS' (Computer Accuracy Precision Score) algorithm.

Moves are still compared against the top engine recommendations, but the math on how these are calculated has changed. Why? Well, most chess players - even low-rated ones - make a lot of the best moves!

Previously, CAPS (v1) looked to create a 0-100 band within the normal human player range. So, scoring perhaps 40% 'Best' moves, which is very low, was equal to single digits on CAPS. And, scoring a very high number of 'Best' moves, but not a perfect game, was often rated 99.9 on CAPS, even though it wasn't played perfectly.

This made some people feel bad (on the low end), and led to a lot of cheating allegations (on the high end). **The new Accuracy scores, based on CAPS2, replicate the feeling of being graded on a test in school.**

You will notice that the majority of scores now fall mostly between 50 and 95, which provides a more intuitive understanding of how accurately you played in your game.

The image below shows Accuracy scores from CAPS and CAPS2 for players rated between 1000 and 1500.

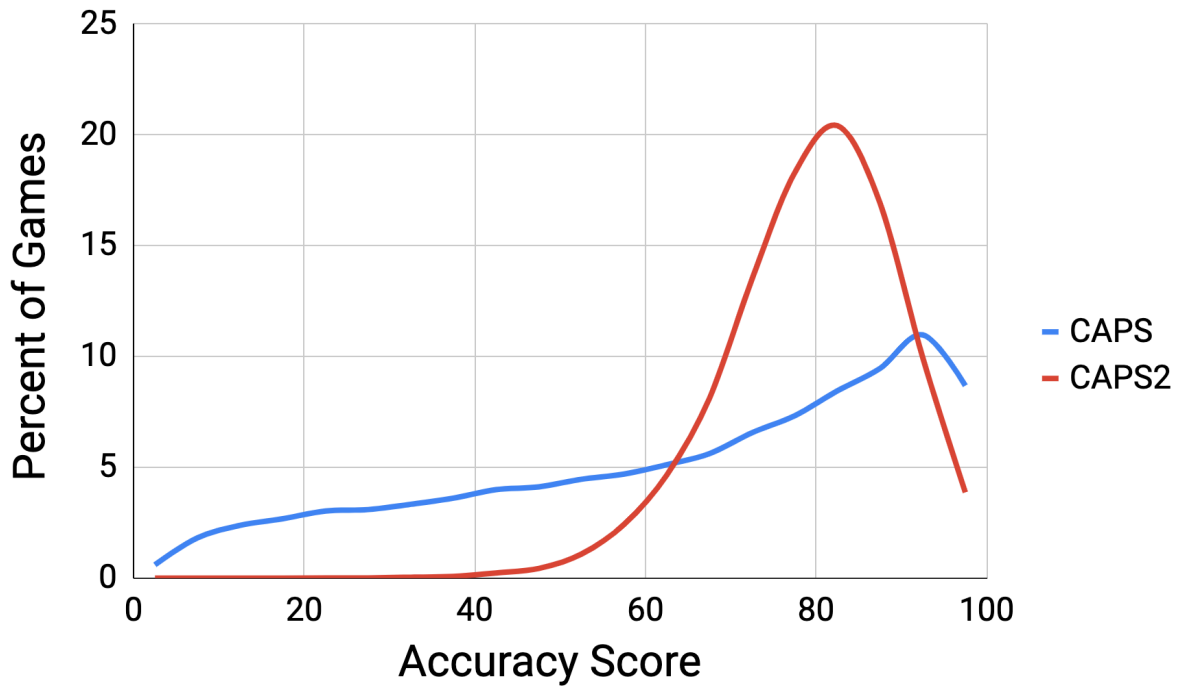


Figure 1: Accuracy scores for players rated between 1000 and 1500.

📰 Check out this article to learn more about Game Review: How does Game Review work?

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